Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Almeida, Eduardo A C Ph.D.		
Project Title:	Does Gravity Influence Matrix Survival Signaling		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	E.Almeida@nasa.gov	**Fax:**	FY 650-604-3159
PI Organization Type:	NASA CENTER	**Phone:**	650-604-1772
Organization Name:	NASA Ames Research Center		
PI Address 1:	MS 236-7		
PI Address 2:	Building 236 Room 217		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	At 1G cell survival and growth are regulated by several environmental signals, including mechanical interactions with extracellular matrix (ECM), other cells, and soluble growth and survival factors. We hypothesize that gravity is a component of the mechanical environment needed for the efficient transduction of cell growth and survival signals from the ECM and that reduction of forces generated by gravity, may increase programmed cell death (apoptosis) and significantly contribute to the musculoskeletal tissue atrophy observed in response to microgravity. The aims of the work: We propose are first to determine if microgravity and hypergravity regulate matrix survival signaling and second to define how known cell survival signaling pathways are affected by microgravity and hypergravity. In initial tests of our hypothesis, we found that simulating increased gravitational acceleration by centrifugation		

Task Description:	enhanced survival of primary fibroblasts and osteoblasts, under conditions where cells depend exclusively on ECM for survival. Centrifugation also stimulated the formation of focal contacts and enhanced activation of FAK, which is required for transduction of survival signals from the ECM. Finally, randomizing the direction of the gravity vector using the Rotating Wall Vessel culture system, a strategy for simulating microgravity, stimulated apoptosis. These results suggest that matrix survival signaling pathways can be modulated by forces resulting from gravity. As a result of these investigations we expect to significantly enhance our understanding of how matrix survival signaling pathways are affected by variations in gravity, and start devising new strategies to counter apoptosis-related effects of microgravity in musculoskeletal tissue atrophy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal addresses fundamental questions about how the gravity environment impacts survival and mechanotransduction in primary cell cultures that model both load-bearing (bone) and non load-bearing connective (dermal) tissues: osteoblasts and dermal fibroblasts. Since osteoblasts play a critical role in skeletal turnover, and contribute to the relative decline in bone mass caused by spaceflight (Whedon, 1994, Morey-Holton, 1994, Morey-Holton and Globus, 1998), results from this study will provide novel insights into potential molecular mechanisms for disuse osteoporosis caused by weightlessness. We will also identify specific signaling pathways needed to maintain cell survival in altered gravity environments. In contrast, our studies of fibroblasts will address the important question of whether gravity effects on cells are exclusively associated with load-bearing tissues such as bone, or if this is a more general phenomenon that can also be observed in non load-bearing connective tissues like skin. In addition, analysis of dermal fibroblasts will provide continuity and a wealth of valuable comparisons with our previous studies of fibronectin/integrin/FAK survival signaling, in primary fibroblasts. In our proposed studies both fibroblasts and osteoblasts will be derived from primary tissue and used directly (osteoblasts) or in the first few passages after isolation (dermal fibroblasts). Since both cell types are primary cultures, they are uniquely relevant models to ask specific questions about cell survival, which would be difficult to address in transformed cell lines that often have lost anchorage dependence and the ability to undergo ECM-dependent apoptosis. Since most cell types depend on adhesion to the ECM for survival, we expect that what we learn from studying the survival of fibroblasts and osteoblasts will be relevant to many different cell and tissue types. Furthermore, these are significant studies because they address a fundamental biological question about the molecular role of gravity-generated forces in regulating cell and tissue survival. In the studies we propose, we will use both cell culture centrifugation and the Rotating Wall Vessel technologies to analyze how the gravity environment affects cell survival. The use of those methods, which have emerged as important tools for research in tissue engineering (Unsworth and Lelkes, PI, 1998), will also enhance our understanding of the influence of existing ground-based technologies for manipulating the gravity vector. Finally tissue loss during spaceflight is one of the most vexing problems challenging future human exploration and development of space. Understanding the molecular basis of this problem is an important priority and it might allow us to gain insight into possible avenues for therapeutic intervention in various forms of microgravity tissue atrophy and loss. Specifically we might be able to devise novel ways to prevent apoptosis in microgravity, possibly by activating pharmacologically compensatory pro-survival kinase signaling pathways. After this project is completed results from these ground-based experiments will provide the basis for the rational design of future spaceflight experiments, that more directly address the role of microgravity in fundamental cell survival signaling pathways.
Task Progress:	No progress provided this reporting period.
Bibliography Type:	Description: (Last Updated: 12/08/2025)